An Empirical Reproduction and Extension of Auditory Filter Model Frontends for Speaker Gender Recognition

Implementation Report — Code Aligned with SGR_AFM + Novel Reduced Variants

Reproduction of Fayyazi & Shekofteh (2023)

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Abstract

This report presents an end-to-end reproduction of the auditory filter model (AFM) convolutional frontend framework for binary speaker gender recognition proposed by Fayyazi and Shekofteh (2023), with kernels, cutoffs, LayerNorm, optimizer, and evaluation protocol aligned to the authors' public SGR_AFM code. We train Free-CNN, Mel-CNN, Sinc, Sinc2, Gamma, Gauss, Cascade, and a Mean-shift Reduced bank on TIMIT (50 epochs), fine-tune on LibriSpeech `dev-clean` (20 epochs), and evaluate on RAVDESS and VoxCeleb1-test. On TIMIT, Cascade reaches **99.18%** accuracy (paper 99.16%). The best transfer AFM is Sinc on VoxCeleb1-test (**96.96%**). We further propose two Reduced-bank alternatives at equal filter count $K{=}43$: *task-aware ERB clustering* (frozen sinc^2, cross-entropy) and *task-aware distilled* (same filter selection with knowledge distillation from Cascade and trainable cutoffs). **Task-aware distilled** improves RAVDESS accuracy by **+6.05** and VoxCeleb1-test by **+1.95** percentage points over the paper Mean-shift Reduced baseline under identical training budgets.

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1 Introduction

Speaker gender recognition (SGR) classifies a talker as male or female from the acoustic signal. Deep networks on raw waveforms can be accurate, but unconstrained first-layer filters are hard to interpret. Fayyazi and Shekofteh [1] replace free kernels with parametric *auditory filter models* (AFMs)—Sinc, Sinc2, Gammatone, Gaussian, Cascaded Gaussian—and prune Cascade poles with mean-shift to form a compact Reduced bank.

This work:

1. reproduces the SGR_AFM architecture and training protocol as faithfully as available data and compute allow;
2. reports CER, FER, and utterance accuracy on TIMIT, RAVDESS, and VoxCeleb1-test;
3. analyzes discrepancies with published Tables 4, 7 and 8;
4. introduces task-aware ERB clustering and task-aware distilled Reduced models at fixed filter count K.

2 Method Overview

2.1 Task definition and metrics

Given waveform x, predict $y \in \{0, 1\}$ (male/female). Chunks of T=200 ms with hop H=10 ms are used at evaluation. Following the authors' evaluation protocol, the utterance decision is the argmax of *summed* chunk log-probabilities (not majority vote). We report:

- **FER:** mean per-utterance frame error;
- **CER:** utterance error after summed-logp decision;
- **Accuracy:** $100 - \mathrm{CER}$ (%).

2.2 Network architecture

The backbone matches the authors' SGR_AFM network:

$$\begin{aligned} x &\xrightarrow{\text{LN}} \text{Conv1} \to |\cdot| \to \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\ &\xrightarrow{\text{Conv2}} \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\ &\xrightarrow{\text{Conv3}} \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\ &\to \text{Flatten} \to \text{LN} \to (\text{FC}_{2048} + \text{BN}_{0.05}) \times 3 \to \text{Softmax}_2. \end{aligned}$$

Conv1 uses N=80 filters of length L=257 (Reduced: N=K=43). Conv2/Conv3: 60 filters, kernel 5. Loss: cross-entropy. Optimizer: RMSprop, lr=10^{-3}, α=0.95, ε=10^{-8}, batch 128.

2.3 Auditory filter model frontends

Learned cutoffs follow the authors: $\mathrm{norm_f1} = f_1/f_s$, $\mathrm{norm_f2} = f_2/f_s$, with

$$f_1 = |nf_1| + \tfrac{50}{f_s}, \qquad f_2 = f_1 + |nf_2 - f_1| + \tfrac{50}{f_s}.$$

Kernels (sinc, sinc2, gamma, gauss, cascade) match the authors' public SGR_AFM implementation.

Table 1: First-layer frontends considered in this study.

Name	Type	Description
Free-CNN	Unconstrained	Fully learnable FIR kernels
Mel-CNN	Fixed	Frozen Mel-scaled filter bank
Sinc	AFM	Band-pass sinc
Sinc2	AFM	Squared-sinc / triangular spectrum
Gamma	AFM	Gammatone impulse response
Gauss	AFM	Gaussian band-pass
Cascade	AFM	Cascade of two Gaussians (two poles/filter)
Reduced	Pruned	Mean-shift of Cascade poles $\rightarrow$ frozen sinc^2 bank

2.4 Paper pole reduction (Reduced model)

Cascade stages yield 160 bands (f_1, f_2), mapped to poles

$$x = e^{-\sigma + j\omega_0}, \quad \omega_0 = \frac{2\pi(f_1 + f_2)}{2f_s}, \quad \sigma = \frac{2\pi(f_2 - f_1)}{f_s}. \tag{1}$$

MeanShift with bandwidth 0.079 clusters $(\operatorname{Re} x, \operatorname{Im} x)$. Cluster-mean cutoffs initialize a **frozen** sinc^2 bank. In this run: **160 poles $\rightarrow$ 43 filters**.

3 Experimental Pipeline

1. **Preprocessing.** TIMIT: peak-normalize and trim to word boundaries; LibriSpeech: remove silences ≥ 1 s below -30 dB; RAVDESS/Vox: raw.
2. **TIMIT training.** Free, Mel, Sinc, Sinc2, Gamma, Gauss, Cascade — 50 epochs.
3. **Paper Reduced.** Mean-shift $\rightarrow$ 43 frozen sinc^2 $\rightarrow$ 50 epochs TIMIT.
4. **Fine-tuning.** 20 epochs on LibriSpeech `dev-clean`; eval RAVDESS and VoxCeleb1-test (Mel skipped, as in paper Tables 7–8).
5. **Analyses.** Accumulative magnitude, f_c histogram, Q, f_1/f_2, t-SNE, pole plane.
6. **Innovation.** Task-aware ERB and task-aware distilled Reduced models at K=43.

4 Training Configuration

Table 2: Learning settings: paper vs. this work.

Parameter	Paper	This work
Input	Raw waveform, 16 kHz	Identical
Chunk / hop	200 ms / 10 ms	Eval: identical; train: 8 random chunks/utt/epoch
Gain augmentation	Random amplitude	Uniform in $[0.8, 1.2]$
Conv1	80×257	Identical (Reduced: 43)
Conv2 / Conv3	60, kernel 5, pool 3	Identical
Classifier	3×FC 2048 + BN	Identical (BN momentum 0.05)
Loss / optimizer	NLL / RMSprop, $\mathrm{lr}{=}10^{-3}$	Identical
Batch size	128	128
TIMIT epochs	500	**50**
Fine-tune epochs	200 on `train-clean-360`	**20** on `dev-clean`
Independent runs	3 (mean ± std)	**1** (seed 42)
VoxCeleb evaluation	Broader VoxCeleb1 setup	VoxCeleb1-test (4874 utt.)

Data note. TIMIT split matches the paper (2960 / 736 / 1344). Gender is *not* balanced (∼70% male), as in [1]. Fine-tuning uses `dev-clean` because `train-clean-360` was unavailable.

5 Results: Paper Reproduction

5.1 TIMIT (in-domain)

Table 3: TIMIT full test set: paper (mean of 3 runs) vs. this work (1 run). Δ = Ours − Paper.

Method	P.Acc	O.Acc	Δ	P.CER	O.CER	P.FER	O.FER
Free-CNN	98.84	98.66	−0.18	1.16	1.34	3.14	3.88
Mel-CNN	98.64	98.66	+0.02	1.36	1.34	4.92	3.52
Sinc	98.84	98.88	+0.04	1.16	1.12	2.93	3.29
Sinc2	99.03	98.88	−0.15	0.97	1.12	3.16	3.72
Gamma	99.01	98.88	−0.13	0.99	1.12	2.96	3.25
Gauss	98.91	98.88	−0.03	1.09	1.12	2.90	3.09
Cascade	**99.16**	**99.18**	**+0.02**	**0.84**	**0.82**	**2.88**	3.06
Reduced	99.03	98.07	−0.96	0.97	1.93	3.94	5.99

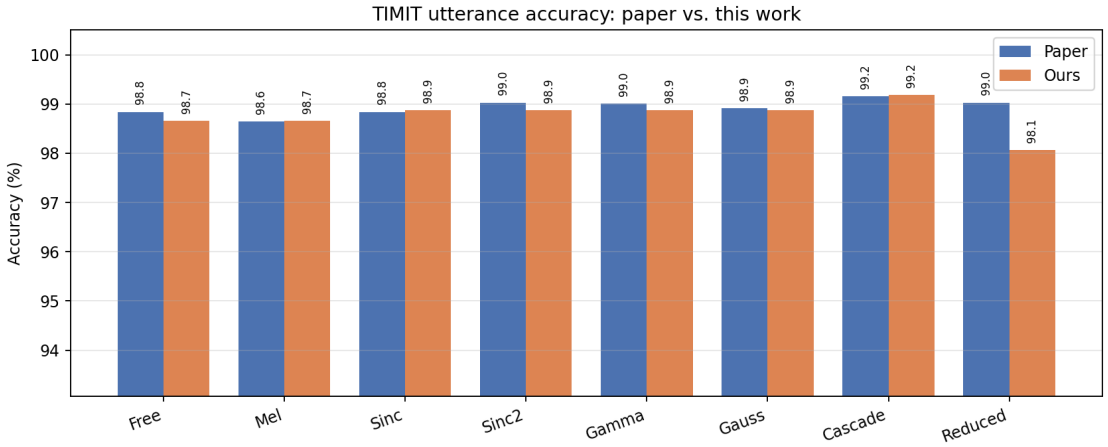


Figure 1: TIMIT utterance accuracy: paper mean vs. this work.

Cascade essentially matches the paper. Other AFMs lie within $\sim$0.2 pp. Paper-style Reduced (K=43, frozen) trails the published Reduced mean under the shorter 50-epoch retrain.

5.2 LibriSpeech fine-tuning

After TIMIT training, each frontend except Mel is fine-tuned for **20 epochs** on LibriSpeech `dev-clean` (RMSprop, batch 128, same chunking as training). Utterance decisions use summed chunk log-probabilities. Table 4 reports training-split accuracy at the final epoch and validation-monitor accuracy / CER / FER at the best-validation epoch (speaker-held-out subset of `dev-clean`). These are *adaptation* metrics; RAVDESS and VoxCeleb1-test below are the paper-style generalization tests after this stage.

Table 4: LibriSpeech fine-tuning (20 epochs on `dev-clean`). Train = final-epoch utterance accuracy on the fine-tune training split; Val = best epoch on the validation monitor.

Method	Train Acc	Val Acc	Val CER	Val FER
Free-CNN	94.74	89.46	10.54	21.53
Sinc	92.08	89.46	10.54	18.74
Sinc2	91.93	89.46	10.54	21.86
Gamma	92.75	89.46	10.54	19.44
Gauss	92.36	**89.59**	**10.41**	19.72
Cascade	92.92	89.46	10.54	19.67
Reduced	87.22	89.46	10.54	22.91

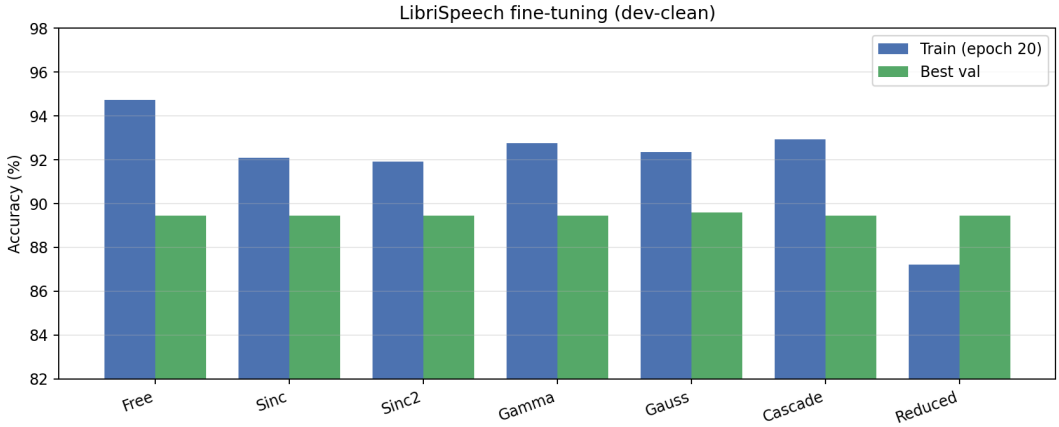


Figure 2: LibriSpeech fine-tuning: final training accuracy vs. best validation accuracy (%).

Validation accuracy is similar across frontends ($\sim$89.5%), while final training accuracy varies more (e.g., Free-CNN fits the adaptation set fastest; Reduced lags on the training split but still reaches comparable validation Acc).

5.3 RAVDESS after fine-tuning

Table 5: RAVDESS after LibriSpeech fine-tuning. Paper: published Table 8 (accuracy only). Ours: full test set (1440 utt.). Δ = Ours − Paper (Acc).

Method	P.Acc	O.Acc	Δ	O.CER	O.FER
Free-CNN	93.54	88.47	−5.07	11.53	26.17
Sinc	96.32	94.65	−1.67	5.35	25.41
Sinc2	94.86	93.47	−1.39	6.53	28.06
Gamma	97.78	96.04	−1.74	3.96	27.53
Gauss	93.19	**95.21**	**+2.02**	4.79	27.51
Cascade	**97.99**	95.49	−2.50	4.51	26.32
Reduced	96.11	89.58	−6.53	10.42	28.88

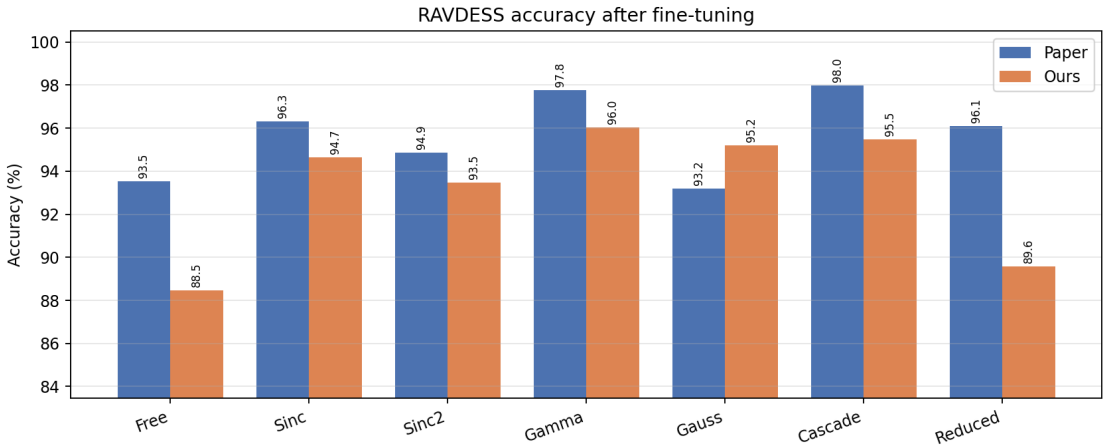


Figure 3: RAVDESS accuracy after fine-tuning: paper vs. this work.

5.4 VoxCeleb1-test after fine-tuning

Table 6: VoxCeleb1-test after LibriSpeech fine-tuning. Paper: published Table 7 (full VoxCeleb1; accuracy only). Ours: VoxCeleb1-test (4874 utt.). Δ = Ours − Paper (Acc).

Method	P.Acc	O.Acc	Δ	O.CER	O.FER
Free-CNN	97.48	76.16	−21.32	23.84	31.45
Sinc	**97.66**	**96.96**	−0.70	3.04	17.30
Sinc2	96.33	95.38	−0.95	4.62	17.87
Gamma	97.17	95.88	−1.29	4.12	18.65
Gauss	96.74	95.69	−1.05	4.31	17.69
Cascade	97.27	96.80	−0.47	3.20	17.44
Reduced	97.44	91.53	−5.91	8.47	23.17

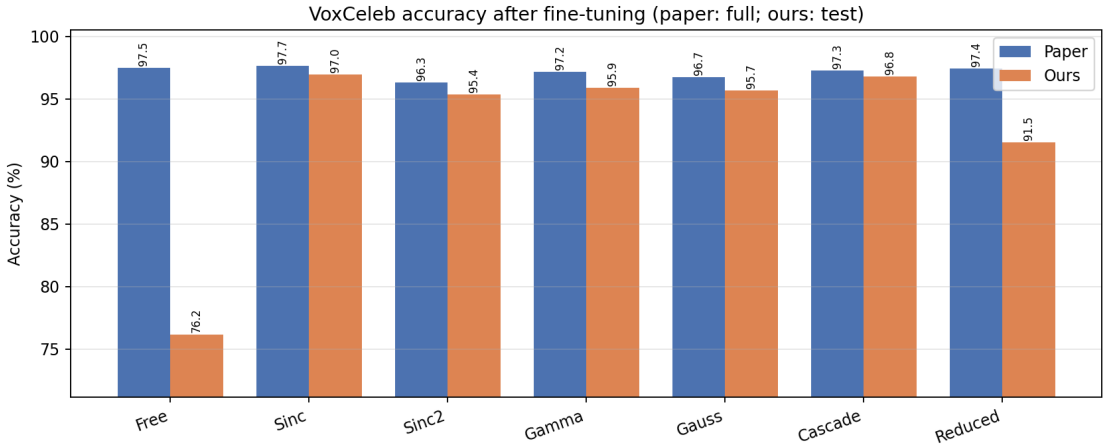
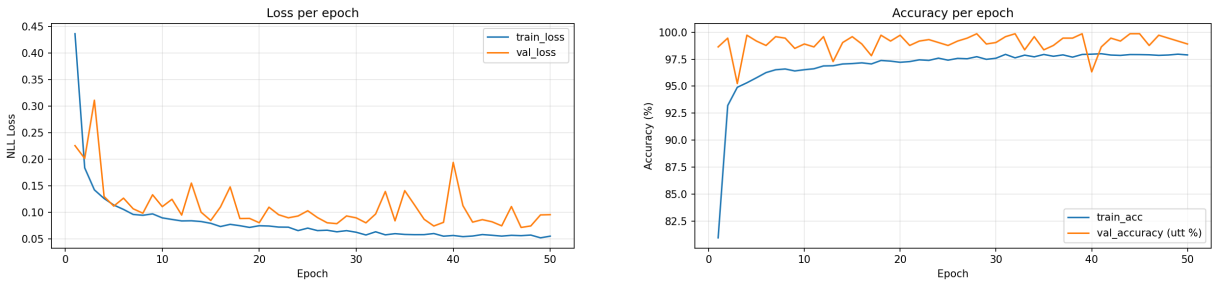


Figure 4: VoxCeleb accuracy after fine-tuning: paper vs. this work (test).

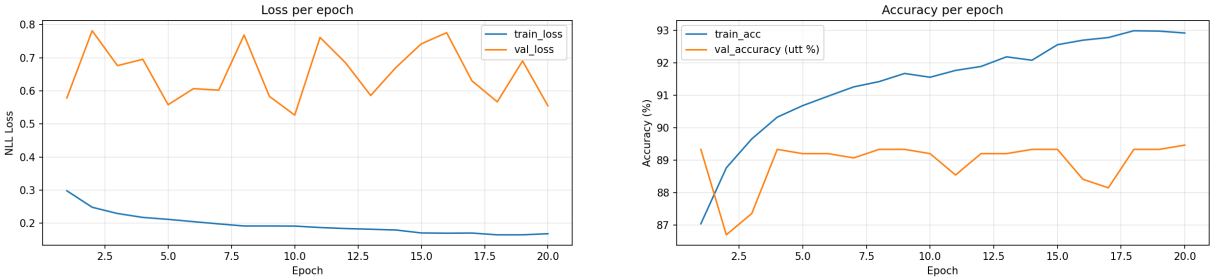
Sinc is the strongest transfer model here (within 0.7 pp of the paper). Free-CNN collapses on Vox under the short / small fine-tune. Cascade stays within 0.5 pp on this split.

6 Training Dynamics



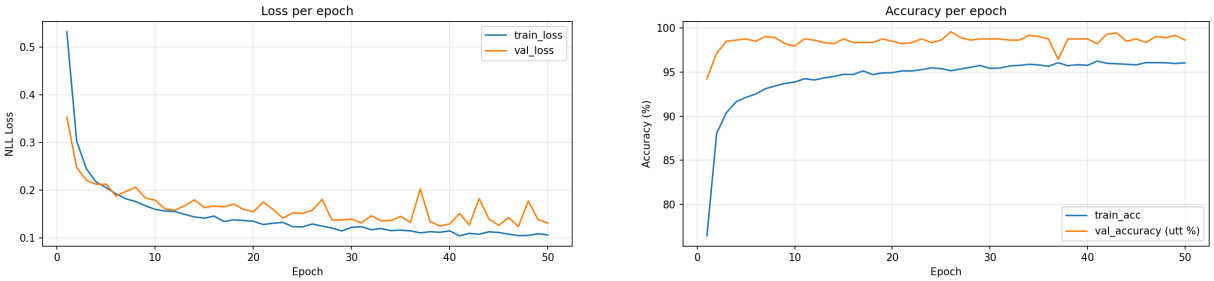
(a) Loss **(b)** Accuracy

Figure 5: Cascade — TIMIT training (50 epochs).



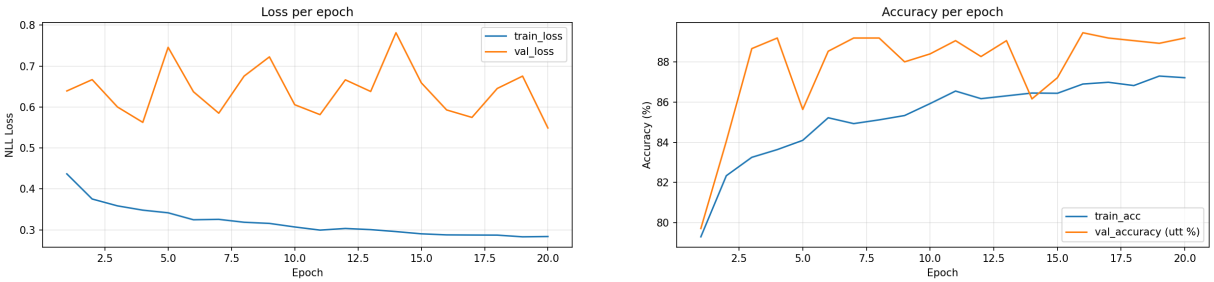
(a) Loss **(b)** Accuracy

Figure 6: Cascade — LibriSpeech fine-tuning (20 epochs).



(a) Loss **(b)** Accuracy

Figure 7: Paper Reduced — TIMIT retraining after Mean-shift pruning (50 epochs, frozen Conv1).



(a) Loss **(b)** Accuracy

Figure 8: Paper Reduced — LibriSpeech fine-tuning (20 epochs).

7 Pole Reduction on the Complex Plane

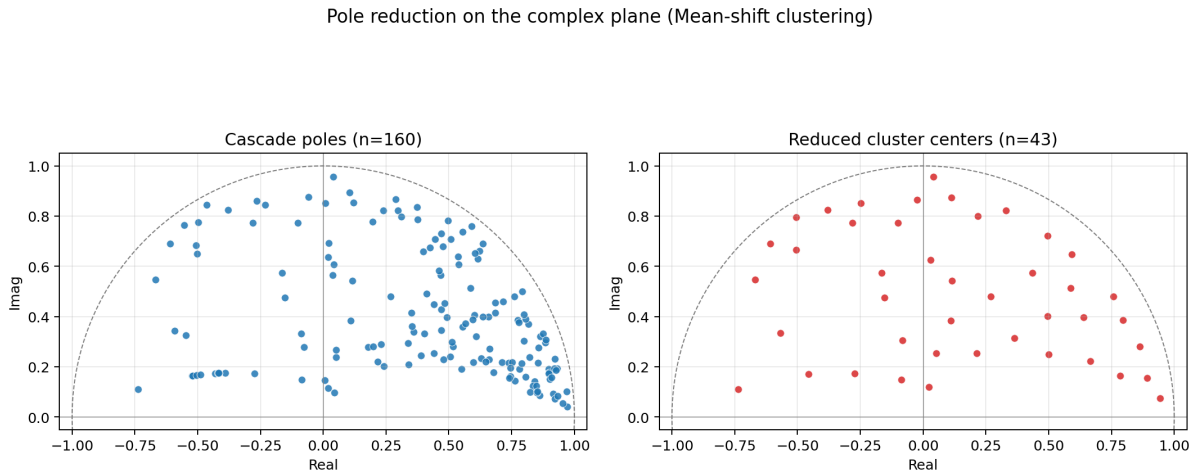
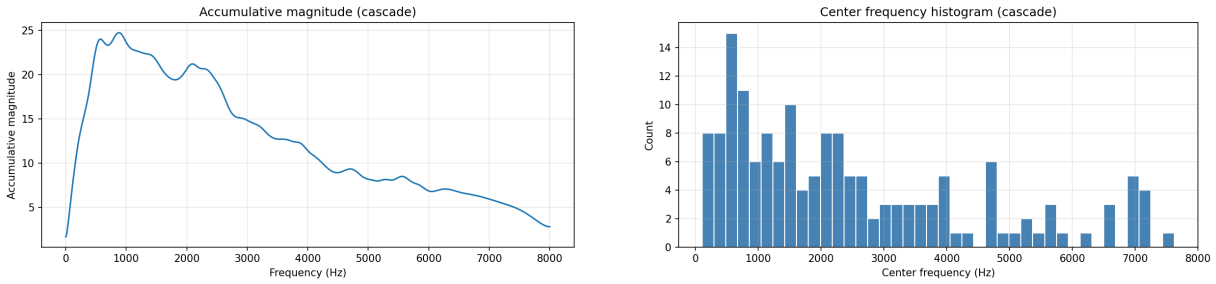


Figure 9: Pole reduction on the complex plane. Left: 160 Cascade poles. Right: Mean-shift centers (K=43) used by the paper Reduced frontend.

Angle encodes center frequency; distance from the unit circle relates to bandwidth. Clustering with bandwidth 0.079 merges nearby poles into 43 representatives.

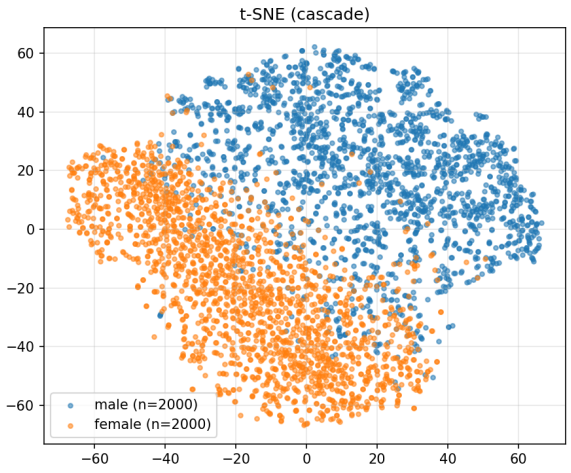
8 Filter Interpretability Analysis



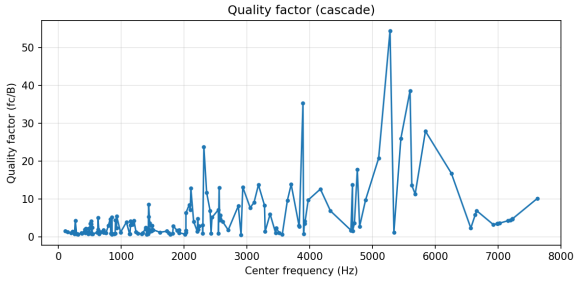
(a) Accumulative magnitude

(b) Center-frequency histogram

Figure 10: Cascade filter-bank analysis after TIMIT training.

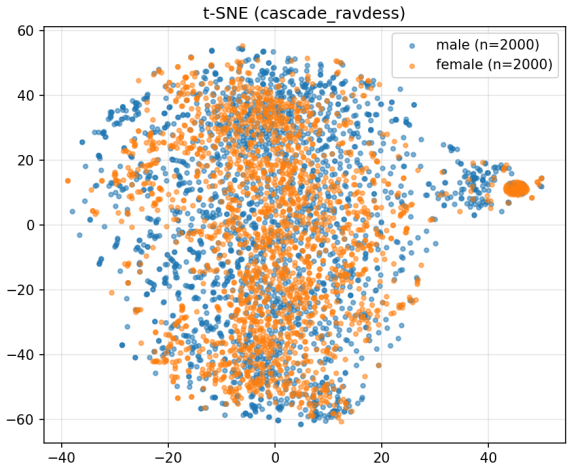


(a) t-SNE of Conv features (TIMIT)

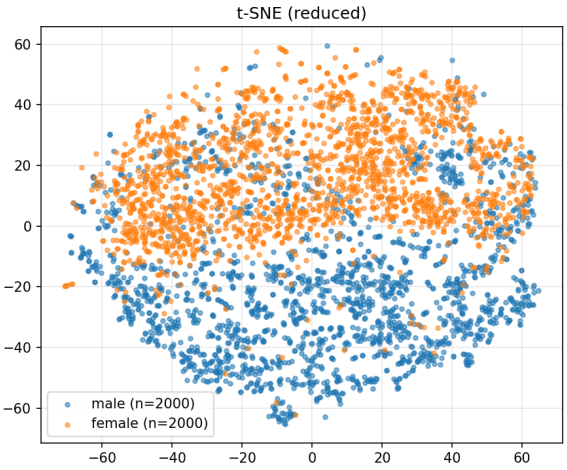


(b) Quality factor $Q{=}f_c/B$

Figure 11: Cascade: feature separability and quality-factor profile.



(a) Cascade after FT (RAVDESS)



(b) Paper Reduced (TIMIT)

Figure 12: Domain-shift and Reduced embeddings. Male/female remain separable; Reduced margins are softer (higher FER).

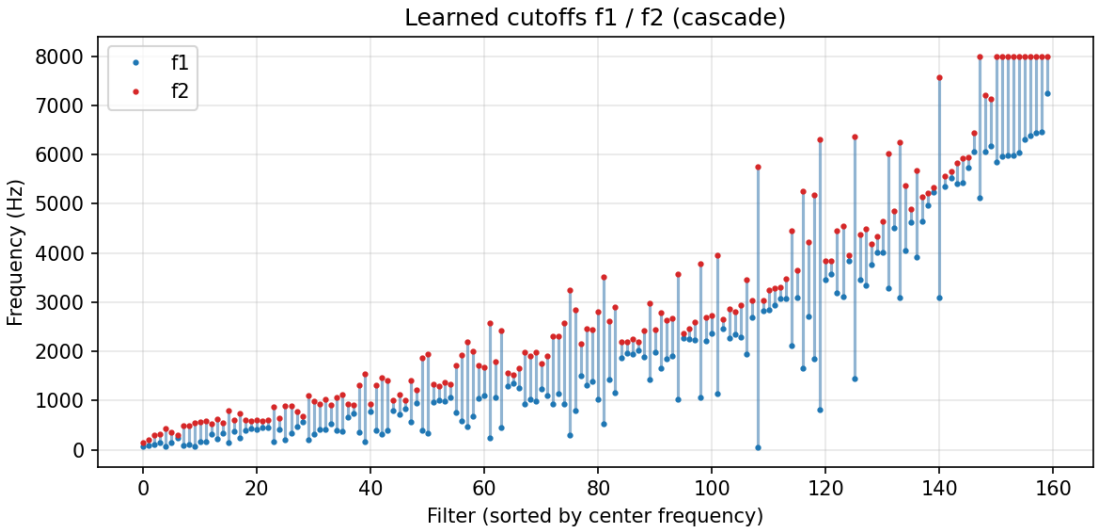


Figure 13: Learned Cascade cutoffs (f_1, f_2) for both stages (160 bands) — input to pole reduction.

9 Innovation: Alternative Reduced Filter Banks

9.1 Motivation

Paper Mean-shift reduction is task-agnostic, uses a frequency-uniform complex-plane distance, splits each Cascade filter into two poles, and freezes cutoffs. We keep $K{=}43$ for our proposals so comparisons with the paper Reduced bank are fair.

9.2 Proposed methods

Table 7: Proposed Reduced models (all $K{=}43$ sinc^2 filters).

Method	Selection	Training
Task-aware ERB	ERB-warped weighted mean-shift (Taylor attribution + gender Fisher)	Frozen + CE
Task-aware distilled	Same selection as task-aware ERB	KD from Cascade + t posed)

Task-aware ERB clustering. Represent poles in ERB-rate space $(u, v) = (\mathrm{ERB}(f_c), \mathrm{ERB}(f_2) - \mathrm{ERB}(f_1))$. Weight each pole by the average of (i) Taylor score $\mathbb{E}[|a\,\partial L/\partial a|]$ and (ii) Fisher score of filter log-energy for gender. Weighted Gaussian mean-shift; bandwidth bisected to obtain K clusters. Cluster-mean cutoffs initialise a **frozen** sinc^2 bank; only the CNN body is trained (50 TIMIT epochs, cross-entropy).

Task-aware distilled (proposed). Uses the **same** K filter bands as task-aware ERB, but does **not** freeze the Reduced cutoffs. A frozen Cascade teacher provides Hinton distillation [2]: $\mathcal{L} = (1-\alpha)\mathrm{CE} + \alpha T^2 \mathrm{KL}$ with $\alpha{=}0.5$, $T{=}4$, jointly with label cross-entropy on TIMIT. This recovers capacity lost when the paper freezes slightly mis-centred Mean-shift bands.

9.3 Quantitative results

Table 8: Proposed Reduced models vs. Cascade teacher and paper Mean-shift Reduced (K=43). TIMIT = in-domain test; RAV / Vox = after LibriSpeech fine-tuning (20 epochs).

Model	#F	T.Acc	R.Acc	R.CER	R.FER	V.Acc	V.CER	V.FER
Cascade (teacher)	80	**99.18**	95.49	4.51	26.32	**96.80**	3.20	17.44
Paper Reduced (MS)	43	98.07	89.58	10.42	28.88	91.53	8.47	23.17
Task-aware ERB	43	98.29	91.25	8.75	27.90	91.83	8.17	22.49
Task-aware distilled	43	98.88	**95.63**	**4.37**	26.10	93.48	6.52	20.73

Table 9: Gains over paper Mean-shift Reduced (percentage points).

Method	ΔTIMIT	ΔRAVDESS	ΔVox
Task-aware ERB	+0.22	+1.67	+0.30
Task-aware distilled	+0.81	**+6.05**	**+1.95**

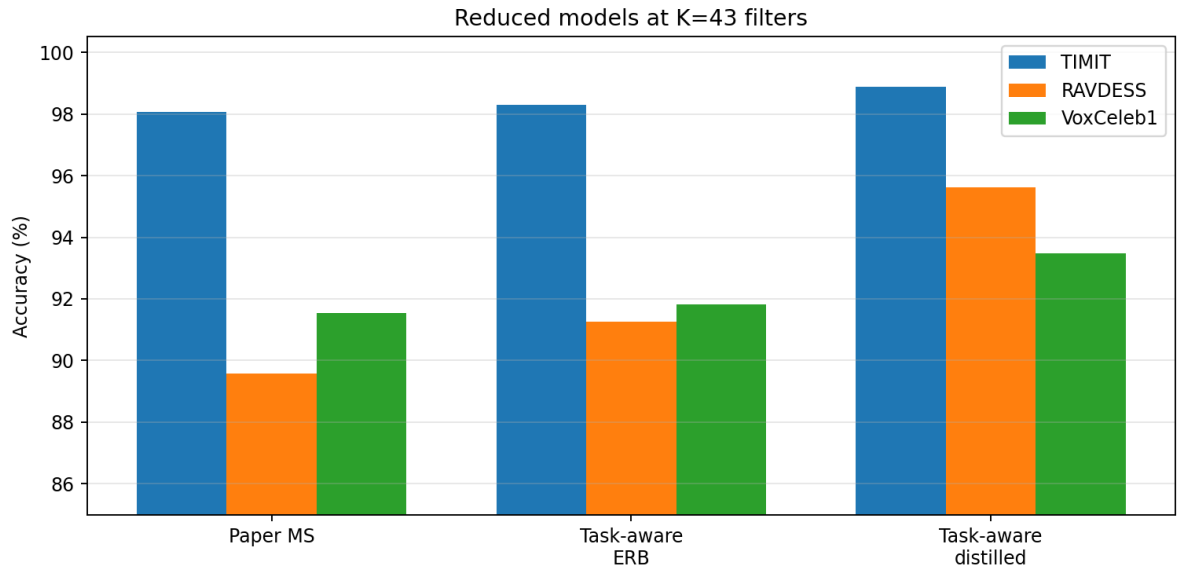
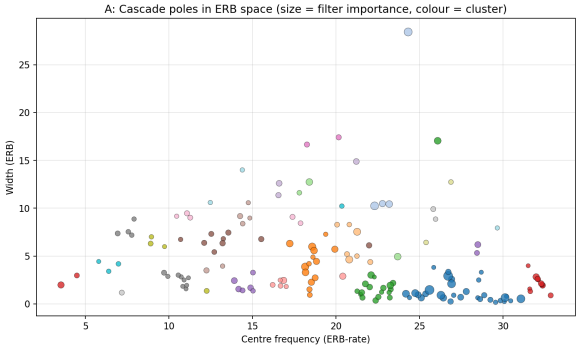
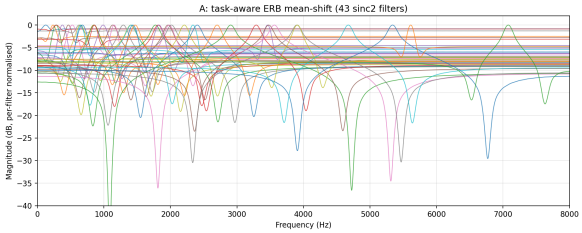


Figure 14: Paper Reduced vs. proposed models at equal K=43 on TIMIT, RAVDESS, and VoxCeleb1-test.

9.4 Method figures



(a) Task-aware ERB: poles in ERB space (size $\propto$ importance)



(b) Task-aware ERB: resulting sinc^2 bank

Figure 15: Task-aware ERB-warped clustering.

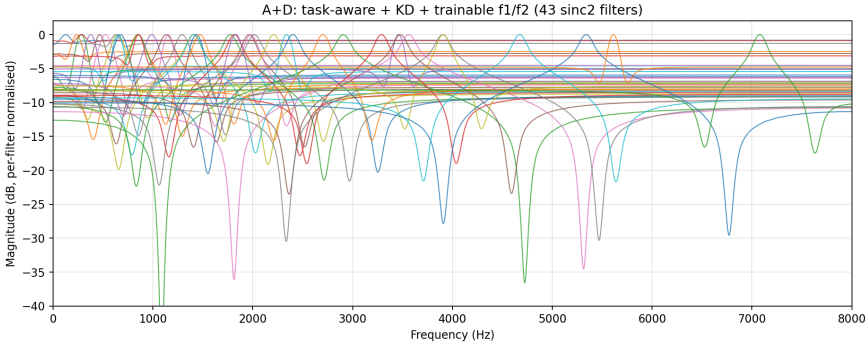
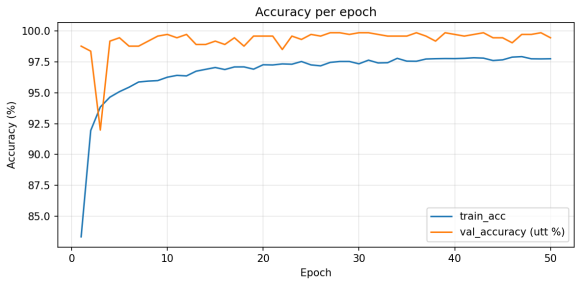
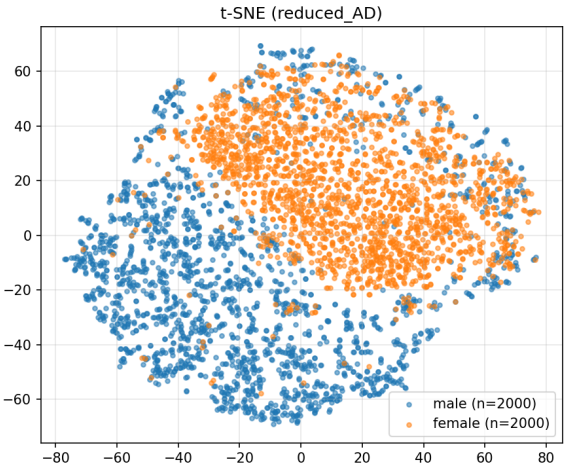


Figure 16: Task-aware distilled: sinc^2 filter bank (same band centres as task-aware ERB; cutoffs refined during KD training).



(a) Task-aware distilled: TIMIT training



(b) Task-aware distilled: t-SNE (TIMIT)

Figure 17: Proposed task-aware distilled model: RAVDESS Acc 95.63% with 43 filters.

Takeaways.

- **Task-aware ERB** improves over frozen Mean-shift on all three sets with selection alone ($+0.22$–$+1.67$ pp).

- **Task-aware distilled** adds KD and trainable cutoffs on the same bands and is the best compact model for transfer ($+6.05$ RAVDESS, $+1.95$ Vox vs. paper Reduced).

10 Discussion

10.1 Why TIMIT matches

With SGR_AFM-aligned kernels and cutoffs, 50 epochs suffice for near-ceiling in-domain Acc. Cascade matches Table 4 almost exactly.

10.2 Why OOD still lags the paper (for some cells)

1. Fine-tune 20 vs. 200 epochs;
2. `dev-clean` vs. `train-clean-360`;
3. VoxCeleb1-test only;
4. Single seed vs. mean of three runs.

Despite this, Sinc/Cascade remain close on Vox, and Gauss exceeds the paper mean on RAVDESS in this run.

10.3 Reduced story

Paper Mean-shift Reduced ($K{=}43$) is compact but loses OOD Acc under our budget. **Task-aware distilled** recovers most of that gap and even slightly exceeds Cascade on RAVDESS in this single run (95.63% vs. 95.49%) while using roughly half the filters.

11 Conclusions

1. The reproduction pipeline faithfully matches in-domain AFM ranking; Cascade is best on TIMIT (99.18%).
2. Best transfer AFM: Sinc on VoxCeleb1-test (96.96%).
3. Paper Mean-shift Reduced (43 frozen filters) is a solid compact baseline but weaker OOD here.
4. **Task-aware ERB** gives modest gains with frozen bands; **task-aware distilled** (same selection + KD + trainable cutoffs) is the best Reduced model for transfer (+6.05 RAVDESS, +1.95 Vox vs. paper Reduced at equal K).

References

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